

LendPerfect Banking App

Project Overview

LendPerfect Banking App is a microservices-based banking and loan management application developed using Java, Spring Boot, Angular, and MySQL. The system helps banks manage customer accounts, loan processing, EMI tracking, transaction management, and secure authentication.

Tech Stack

Backend

- Java 8
- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- REST API
- Microservices

Frontend

- Angular
- TypeScript
- Bootstrap
- HTML/CSS

Database

- MySQL

DevOps & Tools

- Git & GitHub
 - Jenkins
 - Docker
 - AWS
 - Datadog
-

Project Structure

```
lendperfect-banking-app/  
|
```

```
├─ src/main/java/com/lendperfect
│   ├── controller
│   ├── service
│   ├── repository
│   ├── entity
│   ├── dto
│   └── config
├─ src/main/resources
│   └── application.properties
├─ pom.xml
└─ README.md
```

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.lendperfect</groupId>
  <artifactId>lendperfect-banking-app</artifactId>
  <version>1.0.0</version>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-data-jpa</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-security</artifactId>
    </dependency>

    <dependency>
      <groupId>mysql</groupId>
      <artifactId>mysql-connector-java</artifactId>
    </dependency>

  </dependencies>
</project>
```

```
</dependencies>  
</project>
```

application.properties

```
server.port=8080  
  
spring.datasource.url=jdbc:mysql://localhost:3306/lendperfect  
spring.datasource.username=root  
spring.datasource.password=root  
  
spring.jpa.hibernate.ddl-auto=update  
spring.jpa.show-sql=true
```

Customer Entity

```
package com.lendperfect.entity;  
  
import jakarta.persistence.*;  
  
@Entity  
public class Customer {  
  
    @Id  
    @GeneratedValue(strategy = GenerationType.IDENTITY)  
    private Long id;  
  
    private String name;  
    private String email;  
    private String accountNumber;  
    private Double balance;  
  
    public Long getId() {  
        return id;  
    }  
  
    public void setId(Long id) {  
        this.id = id;  
    }  
  
    public String getName() {  
        return name;  
    }  
}
```

```

    public void setName(String name) {
        this.name = name;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    public String getAccountNumber() {
        return accountNumber;
    }

    public void setAccountNumber(String accountNumber) {
        this.accountNumber = accountNumber;
    }

    public Double getBalance() {
        return balance;
    }

    public void setBalance(Double balance) {
        this.balance = balance;
    }
}

```

Repository Layer

```

package com.lendperfect.repository;

import com.lendperfect.entity.Customer;
import org.springframework.data.jpa.repository.JpaRepository;

public interface CustomerRepository extends JpaRepository<Customer, Long> {
}

```

Service Layer

```

package com.lendperfect.service;

import com.lendperfect.entity.Customer;

```

```

import com.lendperfect.repository.CustomerRepository;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import java.util.List;

@Service
public class CustomerService {

    @Autowired
    private CustomerRepository repository;

    public Customer saveCustomer(Customer customer) {
        return repository.save(customer);
    }

    public List<Customer> getAllCustomers() {
        return repository.findAll();
    }
}

```

Controller Layer

```

package com.lendperfect.controller;

import com.lendperfect.entity.Customer;
import com.lendperfect.service.CustomerService;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController
@RequestMapping("/customers")
public class CustomerController {

    @Autowired
    private CustomerService service;

    @PostMapping
    public Customer saveCustomer(@RequestBody Customer customer) {
        return service.saveCustomer(customer);
    }

    @GetMapping
    public List<Customer> getAllCustomers() {
        return service.getAllCustomers();
    }
}

```

```
}  
}
```

Main Application

```
package com.lendperfect;  
  
import org.springframework.boot.SpringApplication;  
import org.springframework.boot.autoconfigure.SpringBootApplication;  
  
@SpringBootApplication  
public class LendPerfectApplication {  
  
    public static void main(String[] args) {  
        SpringApplication.run(LendPerfectApplication.class, args);  
    }  
}
```

API Endpoints

Create Customer

`POST /customers`

Request Body:

```
{  
  "name": "Dusmanta",  
  "email": "dusmanta@gmail.com",  
  "accountNumber": "ACC1001",  
  "balance": 50000  
}
```

Get All Customers

`GET /customers`

Features

- Customer Management
 - Loan Processing
 - Transaction Handling
 - REST APIs
 - Secure Authentication
 - Database Integration
 - Microservices Ready
 - Cloud Deployment Support
-

Git Commands

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/yourusername/lendperfect-banking-
app.git
git push -u origin main
```

Short GitHub Description

"LendPerfect Banking App is a Spring Boot microservices-based banking and loan management system with secure REST APIs, customer management, transaction processing, and cloud-ready deployment support."